ARCHIT AGRAWAL

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PROFESSIONAL EXPERIENCE

EdPlus, Arizona State University **06/2026 - Present**
AI Software Engineer *Phoenix, AZ*

- Designed a serverless data lake on S3: a narrow fact table plus Lance payloads queried in-process by DuckDB, so new datasets union without a migration.
- Built schema-inferring ingestion for any table shape, from crosstabs to weighted surveys and UTF-16 files, emitting each decision as a JSON recipe that replays exactly.
- Traced silent row loss in Step Functions runs that reported success and fixed it with partitioned writes and fan-in across 64 parallel lanes.
- Reached 100% join integrity on repeat runs with schema-constrained LLM output and per-row IDs; the prior pipeline silently mislabeled 1 in 8 rows.
- Tracked data lineage with four typed relations (relabel, recode, wave, wave_recoded), so a second labeling pass can never pose as a trend.
- Parallelized data labeling across 64 AWS Step Functions lanes, with a failure circuit breaker to detect incomplete runs.
- Wired metric lint into CI, so a metric that can no longer be computed against stored runs fails the build instead of rotting.
- Blocked double counting in cross-dataset joins by detecting runs that share the same people.
- Reduced publishing time from 19 minutes to 109 seconds for a 51,280-respondent survey; an independent raw-data audit matched all evaluated figures.
- Cut cross-run disagreements from 30,405 to 263 across 411 runs by giving every column one meaning in one semantic model.
- Secured the analytics proof of concept with entitlement-gated row-level access, Bedrock Guardrails deployed through AWS CDK and enforced per-stage spend limits.

MyStage Music Inc. **03/2025 - 06/2026**
Founding AI/ML Engineer *Remote*

- Scaled ingestion to 70,000+ records daily from 650+ venues using Playwright, transactional locking and entity resolution.
- Made ingestion retries safe with deterministic task IDs and atomic create-if-absent writes, preventing duplicate jobs from overwriting in-flight work.
- Built a Cloud Scheduler reprocessing loop that refreshes the most-stale sources first, tracking reprocess time and count per entity.
- Consolidated a 14-Cloud-Function backend into a checkpointed Python/LangGraph workflow on GCP Cloud Run, reducing production service dependencies 64%.
- Lifted dataset accuracy 25% with Gemini entity resolution, scored by LLM-as-judge, over 70,000+ records a day.
- Designed reusable scraping modules so new data sources can be added without changing the orchestration graph.

EdPlus, Arizona State University **09/2023 - 05/2025**
Software Engineer *Phoenix, AZ*

- Reduced transcript-analysis time from four hours to 15 minutes using a Neo4j knowledge graph and validated Cypher queries.
- Implemented tenant isolation in retrieval services and configuration-based onboarding for additional colleges.

Knowledge Exchange for Resilience, Arizona State University **06/2024 - 08/2024**
Data Research Aide *Phoenix, AZ*

- Built an idempotent ETL pipeline for 10,000+ researcher profiles daily, using PostgreSQL natural-key upserts to prevent duplicates after partial failures.
- Put data-quality gates in PostgreSQL, with uniqueness, not-null and range constraints, so bad rows fail at ingestion; dataset accuracy rose to 95%.

- Cut API latency from 198 to 20 ms with Redis caching and SQL rewrites, verified under simulated peak load with Locust.
- Reduced p95 retrieval latency 60% at 95% recall using hybrid FAISS and Neo4j search.

Zeus Learning **01/2022 - 07/2023**
Software Engineer *Mumbai, India*

- Migrated a C#/.NET monolith to Dockerized Kubernetes microservices, reducing infrastructure cost 20% and resource footprint 35%.
- Improved student-listing performance 30% with pagination and S3 asset delivery; reduced class-detail latency 10% through SQL query rewrites.

EAT.FIT **09/2021 - 12/2021**
Product Intern *Bengaluru, India*

- Automated competitor-catalog harvesting with a Python scraping toolchain that loads catalogs and reviews into the analytics warehouse on a schedule, normalized to one schema.

PROJECTS

Clash Royale Analytics **2025**

- Built a scheduled ETL pipeline that dedupes battle data, loads incrementally and runs backfills without blocking the live cron, with Vitest coverage per stage.
- Built a serverless analytics platform on Cloudflare Workers and D1 with Drizzle ORM, fronted by Next.js 16 and React 19 on Vercel, with cross-filtered D3 visualizations.

Serverless Document Pipeline **03/2024 - 04/2024**

- Built a serverless document pipeline on AWS Lambda, API Gateway, S3 and DynamoDB with a React frontend, handling 1,000+ concurrent requests at 99.9% uptime.

Image Recognition as a Service **01/2024 - 02/2024**

- Built an AWS EC2/SQS image-processing service with queue-based scaling, asynchronous results and idempotent workers for safe message retries.
- Capped worker capacity and used scale-in cooldowns to control cost and prevent instance churn during changing demand.

TECHNICAL SKILLS

Languages: Python, SQL, TypeScript, C#

Storage & Query: S3 data lake, DuckDB, Lance, PostgreSQL, MySQL, DynamoDB, Firestore, SQLite, Redis, Neo4j, pgvector, Algolia

Pipelines & Orchestration: ETL, batch and incremental loads, backfills, AWS Step Functions (Distributed Map), Lambda, SQS, LangGraph, GCP Cloud Run Jobs, Cloud Scheduler, Cloudflare Workers, Playwright

Data Modeling & Quality: Data modeling, fact tables, data warehousing, data validation, constraints, data lineage, schema inference, replayable runs, deduplication, entity resolution

Governance & Reliability: Row-level access control, idempotent reprocessing, circuit breakers, OpenTelemetry tracing

DevOps: Docker, Kubernetes, GitHub Actions, CI/CD, pytest, Vitest

EDUCATION

Arizona State University (ASU) **08/2023 - 05/2025**
M.S., Computer Science | GPA: 4.0 *Tempe, AZ*

Dhirubhai Ambani Institute of Information and Communication Technology **08/2018 - 05/2022**
B.Tech., Information and Communication Technology *Gandhinagar, India*